DSC-002: Mathematical Methods

Lecture 3: **Foundation of Linear Algebra**

Sugata Bag

Delhi School of Economics

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Objectives -

The objective here is to generalize the notion we see on a plane:

- a way to add vectors, and
- to scale them up or down using scalars,
- s.t. vector addition and multiplication with scalars follow certain rules.
- Our scalars will mostly be real numbers; more generally, we want the set of scalars to comprise a *field*.
- We have already learned basics of logic, and group theory
- Let's now get into vectors

Vector Spaces -

Definition

A **vector space** V over a field F of scalars, is a set with 2 operations:
◦ vector addition (called, an **internal composition**) and
◦ multiplication by scalar (called, an **external composition**) s.t. -

(i) V with $+$ is a commutative group with identity element $\overrightarrow{0} \in V$

(ii) If $v \in V$, $cv \in V, \forall c \in F$; and for $c_1, c_2 \in F$, $(c_1 c_2)v = c_1(c_2 v)$;
$1 \in F$, the multiplication identity element for field F, satisfies $1v = v, \forall v \in V$.

(iii) For all $c \in F$ and $v_1, v_2 \in V$, $c(v_1 + v_2) = cv_1 + cv_2$, and
for all $c_1, c_2 \in F$ and $v \in V$, $(c_1 + c_2)v = c_1 v + c_2 v$.

Note that we use the same symbol $+$ for both vector addition and scalar addition but they are different things.

Examples, Pictures

Example

(1). $\mathbb{R}^n$ over the field $\mathbb{R}$, with **vector addition** defined as follows:

$$(x_1, ..., x_n)^T + (y_1, ..., y_n)^T = (x_1 + y_1, ..., x_n + y_n)^T.$$

And **scalar multiplication** as - $c(x_1, ..., x_n)^T = (cx_1, ..., cx_n)^T$.

- Notice geometric interpretation (parallelogram law etc.).
- $1 \in \mathbb{R}$ is the scalar-vector multiplication identity.

(2). $\mathbb{C}^n$ over $\mathbb{C}$.

- Interpret c, x_i, y_i in above **(1)** as complex numbers, and addition and multiplication as that of complex numbers.
- Note that $1 + 0i$ is the scalar-vector multiplicative identity.

(3). $\mathbb{C}[a, b]$ over $\mathbb{R}$, where $[a, b] \subseteq \mathbb{R}$ and $\mathbb{C}[a, b]$ is the set of all real-valued, continuous functions with domain $[a, b]$. [Draw Picture]

Check if the properties of a vector space are satisfied for these examples.

Vector spaces are also called linear spaces. Vectors are elements of a linear space.

Linear Combination and Linear In/dependence

Definition (**Linear Combination**)

Let $v_1, ..., v_m \in A$. A **linear combination** of these vectors is a vector of the form $c_1 v_1 + ... + c_m v_m$, where $c_1, ..., c_m \in F$ field of scalars.

Definition (**Linear Dependence**)

A set of vectors $\{v_1, ..., v_n\}$ over a field of scalars $\{c_1, ..., c_n\}$, **not all zero**, s.t. $c_1 v_1 + ... + c_n v_n = \overrightarrow{0}$ or, $\sum_{i=1}^{n} c_i v_i = \overrightarrow{0}$.
We say, **set of** v_i's are linearly dependent (LD).

[after a while we'd drop the arrow over 0]

Definition (Linear Independence)

If there does **not** exist any such set of not-all-zero scalars, we say $\{v_1, ..., v_n\}$ are **linearly independent** (LI). Alternatively,

. $\{v_1, ..., v_n\}$ is LI if $\sum_{i=1}^{n} c_i v_i = 0 \Rightarrow c_i = 0, \forall i \in \{1, ..., n\}$.

LD and LI - Geometry, Example

(i) What is the geometry in $\mathbb{R}^2$?

2 vectors lying or not lying on the same line.

(ii) To check if $\{v_1, ..., v_n\}$ is LD in say $\mathbb{R}^n$,

- write each $v_i = (a_{1i}, ..., a_{ni})^T$ in terms of its coordinates;
- then we have to check if the system of eqns $Ac = \overrightarrow{0}$ has a non-zero solution $c = (c_1, ..., c_n)^T$ (*co-ordinate vector*).
  Here, A is $n \times n$ matrix whose i^{th} column is vector v_i.
- **Ex:** check if $v_1 := (1, 2, 3)^T$ & $v_2 := (2, 4, 5)^T$ are LD!!

(iii) Any set of vectors that includes the $\overrightarrow{0}$ vector is LD.
Indeed, just multiply it by $c \neq 0$, and multiply all other vectors in the set by scalar 0.

Span(nig) Set

Definition (**Span**)

Let V be a v.s. over the field F and $A = \{v_1, ..., v_n\} \subset V$.

The (linear) **span** of set A is defined as the set of all linear combinations of finite sets of elements of A. It is denoted by $L(A)$ or $span(A)$.

Thus, $L(A) = \{c_1.v_1 + \ldots + c_n.v_n : v_i \in A; c_i \in F, i = 1, \ldots, n\}$.

$L(A)$ is said to be *generated or spanned* by the set A;
and A is said to be the set of generators of $L(A)$.

Clearly, $A \subset L(A)$.

Example

- What's the span of $v_1 = (1, 0)^T$, $v_2 = (1, 1)^T$?
- What's the span of $w_1 = (1, 0, 0)^T$, $w_2 = (0, 0, 1)^T$? Is it entire V^3?

Subspace of a Vector Space V

Definition (**Subspace**)

A non-empty set $S \subseteq V$ is a **subspace** of V, if S itself be a vector space over field F w.r.to the same compositions as defined in V and for all $v_1, v_2 \in S$, $span(\{v_1, v_2\}) \subseteq S$. Therefore, subspace defined as:-

- **(i)** $c.v \in S$ for all $c \in F$ and
- **(ii)** $c_1 v_1 + c_2 v_2 \in S$ for all $c_i \in F$

Fact

1. *A subset S of a vector space V is a sub-space of V if S be closed w.r.to vector addition and multiplication by scalar; i.e. for any $v_1, v_2 \in S$, we have $v_1 + v_2 \in S$ and for scalar $c \in \mathbb{R}$, $c.v_1 \in S$. Therefore, a subspace is itself a vector space*
2. *Every subspace contains $\overrightarrow{0}$ vector (identity element in vector addition). This is true since $0v_1 + 0v_2 = \overrightarrow{0}$ and LHS is a linear combination of v_1, v_2 and hence lies in their span.*

Subspace - Examples

Example

Examples of subspaces of various dimensions in $\mathbb{R}^2, \mathbb{R}^3$ -

- Every line through the origin is a sub-space of Euclidean plane V_2.
- Every plane / line through the origin is a sub-space of Euclidean space V_3.
- What about lines $x \geq 0, y \geq 0$ or a quarter plane $(x, y \geq 0)$ in $\mathbb{R}^2$ or $\mathbb{R}^3$?
- Minimum dimension of subspace in $\mathbb{R}^3$?

Bases - (plural of Basis)

Definition

A set of vectors $\{v_1, ..., v_n\}$ is a **basis** for a vector space V^n if -
(i) it is LI, and **(ii)** $span(\{v_1, ..., v_n\}) = V^n$.

Examples. $\{(1,0)^T, (0,1)^T\}$ and $\{(1,0)^T, (1,1)^T\}$ are 2 different bases for $\mathbb{R}^2$. To establish that, we need to check 2 properties of basis.

(i) **LI**: Is it a case that $c_1(1,0)^T + c_2(1,1)^T = (0,0)^T$ has only zero soln i.e. $c_1 = 0, c_2 = 0$? Yes!!

Note: this is same as finding solns of $Ac = 0$, where $A_{2\times 2}$ is matrix with column vectors $(1,0)^T, (1,1)^T$, & $c = (c_1, c_2)^T$.

(ii) **Span:** Can we write an arbitrary $x \equiv (x_1, x_2) \in \mathbb{R}^2$ as a LC of $(1,0)$ & $(1,1)$? Is there any c_1, c_2 s.t. $(x_1, x_2) = c_1(1,0) + c_2(1,1)$?
Again, we can write this as a matrix eqn $Ac = x$, and solve for $c = (c_1, c_2)$.
So notice the answer turns on the rank of A.

Q. Can $\{(1,0)^T, (0,1)^T, (1,1)^T\}$ form a basis?

Some Comments

- A system consisting of a single non-zero vector is always linearly independent .
- Every super-set of a linearly dependent set of vectors is LD.
- Any sub-set of linearly independent set of vectors is LI.
- The sub-set containing the single element $(1, 0, 0)$ of the space V^3 in F generates the sub-space which is the totality of the elements of the form $(a, 0, 0)$.
  The sub-set $\{(1, 0, 0), (0, 1, 0)\}$ of V^3 in F generates sub-space which is the totality of elements of the form $(a, b, 0)$.
- Thus the sub-set $\{(1, 0, 0), (0, 1, 0)\}$ of V^3 in F can not generate any vector (a, b, c) so they cannot form a basis in V^3.
  In fact, we need another vector, say $(0, 0, 1)$, along with aforementioned two vectors to span entire V^3 and they become a basis.

Unique Representation

Theorem

Suppose $\{v_1, ..., v_n\}$ *is a basis for a vector space* V*. Then every vector* $v \in V$ *has a unique representation* $v = \sum_{i=1}^{n} c_i v_i$ *in terms of the basis.*

(Note: this leads to unique set of c_i*'s for every* v_i*.)*

Proof.

Suppose 2 sets of scalars $(c_1, ..., c_n)$ and $(c_1^{'}, ..., c_n^{'})$ represent v relative to the basis. Then $v = \sum_1^n c_i v_i = \sum_1^n c_i^{'} v_i$.

So, $\sum_1^n c_i v_i - \sum_1^n c_i^{'} v_i = \sum_1^n (c_i - c_i^{'}) v_i = 0$.

So by LI of the basis vectors, $c_i - c_i^{'} = 0$, or $c_i = c_i^{'}, \forall i \in \{1, ..., n\}$. □

Note that the basis vector v_i has the representation $(0, ..., 0, 1, 0, ..., 0)$ (with 1 in the ith place).

Theorems: Dimension of V

Theorem

- *Any two bases of a finite-dimensional vector space V have the same number of elements.*
- *This common number is called the* **dimension** *of V.*

Numerical Example: Bases in $\mathbb{R}^2$

- The standard basis: $\mathbf{e}_1 = (1, 0),\ \mathbf{e}_2 = (0, 1)$
- Another basis: $\mathbf{w}_1 = (1, 1),\ \mathbf{w}_2 = (1, -1)$
- **Observation:**
  - Both sets are linearly independent and span $\mathbb{R}^2$.
  - Can we find a basis with 3 vectors in $\mathbb{R}^2$?
  - **No:** Any set of 3 vectors in $\mathbb{R}^2$ must be linearly dependent.

◦ Every basis of $\mathbb{R}^2$ has exactly 2 vectors.

Theorem: Dimension of V

Proof Idea (Conceptual Outline)

- All bases are maximal LI sets (and minimal spanning sets).
- **Exchange Lemma (Steinitz):**
  If you have a set of n linearly independent vectors and a spanning set of m vectors, then $n \leq m$.
- Apply to two bases:
  - Let one basis have n vectors, the other m.
  - One spans, one is linearly independent $\implies n \leq m$.
  - Swap roles $\implies m \leq n$.
  - So $n = m$.

Proof Sktech: Let $B = \{v_1, \ldots, v_n\}$ and $C = \{w_1, \ldots, w_m\}$ be two bases

- Since B is linearly independent and C spans V, by the Exchange Lemma, $n \leq m$.
- Likewise, since C is linearly independent and B spans V, $m \leq n$.
- Therefore, $n = m$.

Dimension (of V)

Theorem

If $\{v_1, ..., v_n\}$ *and* $\{w_1, ..., w_m\}$ *are bases for a v.s.* V*, then* $m = n$*.*

Proof.

Suppose WLOG $m \geq n$. Since the v's form a basis, $w_1 = \sum_1^n c_i v_i$, with at least one $c_i \neq 0$ (since $w_1 \neq 0$ by LI).

WLOG let $c_1 \neq 0$; then $v_1 = c_1^{-1} w_1 - \sum_2^n c_1^{-1} c_i v_i$. So $\{w_1, v_2, ..., v_n\}$ spans V. It's also LI, as $\{v_2, ..., v_n\}$ are LI and w_1 is not a LC of them (if it were, c_1 would be 0). So $\{w_1, v_2, ..., v_n\}$ is a basis.

We can iterate this argument n times (see below), replacing v_2 with w_2 and so on. After n replacements, we get that $\{w_1, ..., w_n\}$ is a basis. But then $\{w_1, ..., w_m\}$ cannot have any additional vectors: since this is a LI set, an additional vector, say w_{n+1} cannot be a LC of $\{w_1, ..., w_n\}$ contradicting that this latter set is a basis. So, $m = n$. □

Notes

- Note. To expand on the proof, we have a basis $\{w_1, ..., w_{k-1}, v_k, ..., v_n\}$ after $k-1$ iterations.

  So, $w_k = \sum_1^{k-1} a_i w_i + \sum_k^n c_i v_i$, with not all c_i's being 0, because otherwise we would imply that the w_i's were LD.

  WLOG, let $c_k \neq 0$ (by possibly renumbering the vectors $v_k, ..., v_n$), and so v_k is a LC of $\{w_1, ..., w_k, v_{k+1}, ..., v_n\}$ and we can argue this latter set is a basis.

- Note also that in the first iteration, $\{w_1, v_2, ..., v_n\}$, you can show LI more painfully but clearly by showing

  (i) $\{v_2, ..., v_n\}$ is LI since it is a subset of an LI set;
  (ii) w_1 is not a LC of $\{v_2, ..., v_n\}$;
  (iii) therefore, expanding the set $\{v_2, ..., v_n\}$ with w_1 preserves LI.

Linear Transformations

Definition (Linear Transformations)

Let V, W be two vector spaces. A **linear transformation** $T : V \to W$ is a function satisfying $T(c_1 v_1 + c_2 v_2) = c_1 T(v_1) + c_2 T(v_2)$, for all scalars c_1, c_2 and vectors $v_1, v_2 \in V$.

This is equivalent to T satisfying - **(i)** $T(v_1 + v_2) = T(v_1) + T(v_2)$ and **(ii)** $T(cv) = cT(v), \forall c \in K, v \in V$.

Fact

A matrix $A_{m\times n}$ is a linear transformation. $T : \mathbb{R}^n \to \mathbb{R}^m$, defined by $T(x) = Ax, \forall x \in \mathbb{R}^n$. This function is linear because for all $c_1, c_2 \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$, $A(c_1 x + c_2 y) = c_1 Ax + c_2 Ay$.

In a later section, we will discuss the point that any linear transformation between finite dimensional vector spaces can be "represented" as a matrix by fixing bases in the domain and codomain.

Linear Transformation... 2

Example

Rotation of all points in $\mathbb{R}^2$ by γ radians.

(i) Adding $v_1 + v_2$ then rotating the result equals rotating v_1 and v_2, then adding the result. When we add, then rotate, the whole parallelogram used for addition rotates. When we rotate first, this parallelogram rotates; we then add using it. So in either case, we rotate a parallelogram by γ radians and then use it.

(ii) When we stretch v, then rotate, the whole line - on which both v, cv lie - rotates. Equivalent to rotating the line then stretching v.

Draw graphs. Some Numerical examples.

Range/ Column Space (of T)

Definition (Range)

Let $T : V \to W$ be a LT. The **range space** of T, denoted $R(T)$, is the range of the function, i.e. $R(T) = \{w \in W | w = T(v) \text{ for some } v \in V\}$.

- In the context of linear transformations, the range is called Range/ Column space because it is a subspace, which is spanned by the columns of transformation matrix A.
- Indeed, let $w_1, w_2 \in R(T)$ and c_1, c_2 be scalars.
  So, $w_1 = T(v_1)$, $w_2 = T(v_2)$, for some vectors $v_1, v_2 \in V$.
  Searching for a preimage for $c_1 w_1 + c_2 w_2$,

$$c_1 w_1 + c_2 w_2 = c_1 T(v_1) + c_2 T(v_2) = T(c_1 v_1 + c_2 v_2),$$

  the last equality since T is linear.

  So, $c_1 w_1 + c_2 w_2$ has preimage $c_1 v_1 + c_2 v_2$, and is hence in $R(T)$. So $R(T)$ is a subspace.

Range Space

Range spaces of matrices. Let $A_{m\times n} = (\mathbf{a_1}\ \mathbf{a_2}\ \cdots\ \mathbf{a_n})$ written in terms of its n columns. Since $Ax = x_1\mathbf{a_1} + x_2\mathbf{a_2} + ... + x_n\mathbf{a_n}$, $R(A)$ or set of all images like Ax is equal to $span(\{\mathbf{a_1}, ..., \mathbf{a_n}\})$, the subspace spanned by the columns of A.

Draw range spaces of

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{pmatrix}$$

Dimension of Range Space

- Since $R(T)$ is a subspace, it has a dimension. The dimension of $R(T)$ (denoted $dimR(T)$) is called the **Rank** of T.
- A matrix $A_{m\times n}$ with k LI columns has the property that its columns span a k-dimensional subspace. This dimension therefore corresponds to the notion of $Rank(A)$ being equal to the number of LI columns of A.
- What are the dimensions of range spaces in previous examples?

Null Space / Kernel (of T)

Definition

The **Null Space or Kernel** of an LT $T : V \to W$, denoted $N(T)$, is the subset of vectors in V with image 0 in W; i.e.
$N(T) = \{v \in V | T(v) = \overrightarrow{0}\}$.

- **Q:** What are the Null Spaces of the matrices in the previous slide? We can see that they are all subspaces of the domain. This is true in general.
- Note: If $v_1, v_2 \in N(T)$, we have $T(v_1) = T(v_2) = 0$.
  So for all $c_1, c_2 \in K$, $T(c_1 v_1 + c_2 v_2) = c_1 T(v_1) + c_2 T(v_2) = 0 + 0$.
  So $c_1 v_1 + c_2 v_2 \in N(T)$. i.e. an LC of null vectors $\in N(T)$
- The **dimension** of $N(T)$, $dim(N(T)$, is called the **Nullity** of T.

Some Remarks

Notice in the examples of 2×2 matrices that their Rank and Nullity add up to 2, which is also the dimension of the domain $\mathbb{R}^2$.

More generally, suppose A is an $m \times n$ matrix, and suppose its null space has dimension k. That is, a k-dimensional subspace of vectors in $\mathbb{R}^n$ all map to the origin in $\mathbb{R}^m$. Then there remain $n - k$ dimensions of $\mathbb{R}^n$ free and to map the remaining vectors, we need a space of dimension $n - k$ in $\mathbb{R}^m$.

We will use the following fact::

Let $\{v_1, ..., v_k\}$ be a LI set of vectors in a finite dimensional vector space V. We can extend this to a basis $\{v_1, ..., v_k, v_{k+1}, ..., v_n\}$.

If $span(\{v_1, ..., v_k\}) = V$, we need go no further.

Otherwise, there exists a vector, call it $v_{k+1} \notin span(\{v_1, ..., v_k\}$.

So, $\{v_1, ..., v_k, v_{k+1}\}$ is LI. If it spans V, stop; otherwise iterate the argument.

The expansion of this set must stop at some finite set $\{v_1, ..., v_n\}$, because if it doesn't, the vector space is not finite dimensional. This set therefore will span V; and it is LI by construction. Hence it is a basis for V. The following theorem considers a LT $T : V \to W$, where V, W are FDVS.

The Fundamental Theorem of Linear Algebra

Theorem

Let $T : V \to W$. Then $dim(V) = Rank(T) + Nullity(T)$.

Numerical Examples.

Proof. (not required).

Suppose that $\{v_1, ..., v_k\}$ be a basis for $N(T)$.
We extend this to $\{v_1, ..., v_k, v_{k+1}, ..., v_n\}$, a basis for V in the domain, so that $\dim(V) = n$, and $Nullity(T) = k \leq n$.
Then it suffices to show that $\dim R(T) = n - k$, i.e. ,
$\{w_1, ..., w_{n-k}\} \equiv \{T(v_{k+1}), ..., T(v_n)\}$ is a basis for $R(T)$.

Let's conjecture that $\{T(v_{k+1}), ..., T(v_n)\}$ forms a basis of image T.

□

proof continues...next slide

Proof continues ...

Part (i). Show $span(\{T(v_{k+1}), ..., T(v_n)\}) = R(T)$.
Let $w \in R(T)$, so $w = T(v)$ for some $v \in V$.
In terms of the chosen basis, let -
$v = a_1 v_1 + ... + a_k v_k + a_{k+1} v_{k+1} + ... + a_n v_n$, for some scalars $a_1, ..., a_n$.
So, $w = T(v) = T(\sum_1^k a_i v_i) + T(\sum_{k+1}^n a_i v_i) = \sum_{k+1}^n a_i T(v_i)$

□

Note: the 2nd equality is due to linearity of T; the 3rd as $(\sum_1^k a_i v_i) \in N(T)$ and T is linear. Since w is arbitrary in $R(T)$ and is a LC of $\{T(v_{k+1}), ..., T(v_n)\}$, this set spans $R(T)$

We only now need to show that this list is not redundant; i.e. , that $\{T(v_{k+1}), ..., T(v_n)\}$ are *LI*. We can do this by showing that a LC of these vectors is zero *iff* the coefficient on each vector is zero.

Proof continues ...

Part (ii). Show $\{T(v_{k+1}), ..., T(v_n)\}$ is LI.

Let $c_{k+1}T(v_{k+1}) + ... + c_nT(v_n) = \overrightarrow{0}$, [for scalars $c_{k+1}, ..., c_n$]

$\iff T(c_{k+1}v_{k+1} + ... + c_nv_n) = \overrightarrow{0}$, [by linearity]

$\Rightarrow c_{k+1}v_{k+1} + ... + c_nv_n \in N(T)$, [by defn of nullity]

Representing it in terms of the chosen basis for $N(T)$

$c_{k+1}v_{k+1} + .. + c_nv_n = c_1v_1 + .. + c_kv_k \quad [:= \overrightarrow{0}]$, for some $c_1, .., c_k$

$\Rightarrow c_1v_1 + ... + c_kv_k - c_{k+1}v_{k+1} - ... - c_nv_n = \overrightarrow{0}$.

But, since $\{v_1, ..., v_n\}$ form a basis of V, they should be LI, so all the c_i's equal 0, so that $\{T(v_{k+1}), ..., T(v_n)\}$ is LI and span the range. □

For an $m \times n$ matrix A we have $n = Rank(A) + Nullity(A)$.

Application: Systems of Linear Equations

Given $b = (b_1, ..., b_m)^T \in \mathbb{R}^m$, and $x = (x_1, ..., x_n)^T$ is a solution to the system of linear equations (Note that: $m \lesseqgtr n$)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\ \vdots \quad ..... \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m \end{aligned}$$

if x is a preimage of b under the function $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(v) = Av, \forall v \in \mathbb{R}^n$, where A is the $m \times n$ matrix in the above system. More compactly, we can write the linear system of equations as the equation $Ax = b$.

General Solution Set of Ax=b

$$=\textbf{[Particular Solution of } Ax = b\textbf{] + [General Solution of } Ax = 0\textbf{]}$$

Lemma

Let $A_{m\times n}$ be a real matrix, b a vector in $\mathbb{R}^m$, and $x^{'} \in \mathbb{R}^n$ satisfy $Ax^{'} = b$. Then the set of all solutions to the equation $Ax = b$ (i.e. $\{x \in \mathbb{R}^n | Ax = b\}$) is the set $\{x \in \mathbb{R}^n | x = x^{'} + u,$ for some $u \in N(A)\}$.

Proof.

Suppose x is a general solution of $Ax = b$. Now $x = x^{'} + u$, where $u = x - x^{'}$ such that $Au = A(x - x^{'}) = Ax - Ax^{'} = \overrightarrow{0}$, so $u \in N(A)$. Conversely, suppose $x = x^{'} + u$ where $x^{'}$ is a solution and $u \in N(A)$. Then $Ax = A(x^{'} + u) = Ax^{'} + Au = b + \overrightarrow{0}$, so x is a solution. □

Geometrically, $N(A) = \{u \in \mathbb{R}^n | Au = 0\}$ is a subspace. The above lemma tells the set of solutions to $Ax = b$ can be written as $\{x^{'}\} + N(A)$, (where $x^{'}$ is one solution), a parallel translate of this subspace.

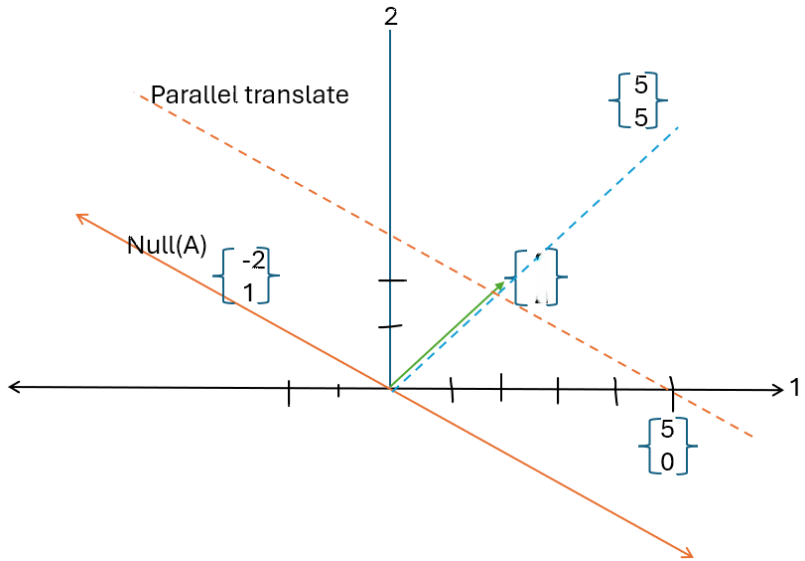
Solution Set. contd ...

Note that if $Nullity(A) \geq 1$, then the system of equations has multiple solutions, if there are any solutions at all.

As an example for the lemma, consider a system of equations with

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} b = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

Draw the solution set and the 1-dimensional subspace $N(A)$.



Linear Equation - Theorems

Theorem (1: Same no. of eqns and variables i.e. $T : \mathbb{R}^n \to \mathbb{R}^n$)

Let A be $n \times n$, and $b \in \mathbb{R}^n$. The system $Ax = b$

(i) *has a* **unique** *solution if $Rank(A) = n$.*

(ii) *may have* **no solution**, *if $Rank(A) < n$.*
But if a solution exists, there are multiple solutions.

Proof.

(i) $Rank(A) = n \Rightarrow R(A) = \mathbb{R}^n$, i.e. A is, safe to assume, surjective [i.e. **onto**, but not necessarily one-one], so a solution (pre-image) exists for every $b \in \mathbb{R}^n$. Moreover, $Nullity(A) = 0$, so $N(A) = \{0\}$, so the solution set (a translate of $N(A)$) has a *single* element [thus, A is one-one; that's why Bijective] – the solution is unique.

(ii) $Rank(A) < n \Rightarrow R(A) \subset \mathbb{R}^n$, so there exists b in codomain with no preimage/soln to eqn system (Is that injective??). Moreover, $Nullity(A) \geq 1$. So if $b \in R(A)$, so that there is a solution (b has a preimage), we have multiple solutions - the translate of a 1 or higher dimensional subspace. □

Linear Equation - Theorems

Theorem (2: Fewer equations than variables i.e. $T : \mathbb{R}^n \to \mathbb{R}^m$)

Let A be $m \times n$, where $m < n$, and let $b \in \mathbb{R}^m$. The system $Ax = b$ then

(i) *has multiple solutions if $Rank(A) = m$.*

(ii) *If $Rank(A) < m$, a solution need not exist; if it does exist, then there are multiple solutions.*

Proof.

(i) $Rank(A) = m \Rightarrow R(A) = \mathbb{R}^m$, since $R(A)$ is then an m-dimensional subspace of $\mathbb{R}^m$. So A is surjective and every $b \in \mathbb{R}^m$ has a preimage. Moreover, $Nullity(A) = n - Rank(A) = n - m > 0$, so the solution set, which is a parallel translate of $N(A)$, holds multiple solutions.
(ii) Proved along similar lines as (ii) of previous theorem. □

NOTE: **(a)** So if A is square and of full rank, the system is '**identified**'; **(b)** if A has fewer rows than columns and is of full rank, the system is '**overidentified**'. **(c)** In the other cases, the system is **not identified**.

Linear Equation - Theorems

Theorem (3: More equations than variables i.e. $T : \mathbb{R}^n \to \mathbb{R}^m, m > n$)

Let A be $m \times n$, $m > n$, and $b \in \mathbb{R}^m$.

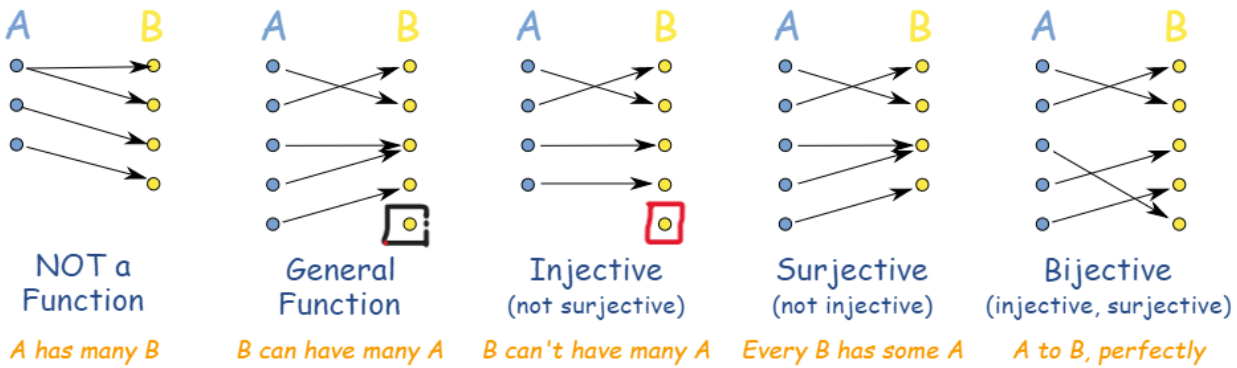
(i) *Then a solution to $Ax = b$ need not exist.*

(ii) *If a solution does exist, it is unique if $Rank(A) = n$ and not unique if $Rank(A) < n$.*

Proof. (Homework).

Mapping - Clarifications

Consider following mappings from elements in domain A to elements in codomain B:



An **onto** (i.e. **Surjective**) mapping covers the entire codomain, ensuring that each element in co-domain has at least one pre-image in the domain.

An **into** (i.e. **Injective**) mapping ensures that each element in the domain maps to a distinct element in the codomain, thus **preventing** any element from having multiple pre-images.

Linear Implicit Function Theorem

Theorem

Consider the system

$$\begin{array}{c} a_{11}x_1 + \ldots + a_{1k}x_k + a_{1k+1}x_{k+1} + \ldots + a_{1n}x_n = b_1 \\ \vdots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \vdots \\ a_{m1}x_1 + \ldots + a_{mk}x_k + a_{mk+1}x_{k+1} + \ldots + a_{mn}x_n = b_m \end{array}$$

Fix $x_{k+1}, \ldots, x_n$ *at values* $x^{\circ}_{k+1}, \ldots, x^{\circ}_n$ *respectively.*
For each such choice of "parameters", there exists a unique set of values $\tilde{x_1}, \ldots, \tilde{x_k}$ *that solves the equation system if and only if*
(i) $k = m$

$$(ii) \qquad Rank \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mk} \end{pmatrix}_{m=k} = k$$

Linear Implicit Function Theorem

Proof.

For each equation i, carry $a_{ik+1}x^{\circ}_{k+1} + ... + a_{in}x^{\circ}_{n}$ to the right hand. Then from the first two theorems above, the only way for a solution to always exist, and be unique, is if the underlying matrix is square and of full rank. □

See Simon and Blume, p 150. The implicit function theorem is a generalization of this and in economics is used to solve for endogeneous variables when one or more parameters change.

Geometry of $Ax = b$ another look:. Look at the ith row vector r_i as a $1 \times n$ matrix. So $r_i x = b_i$ is itself a linear equation. Since $Rank(r_i) = 1$, $Nullity(r_i) = n - 1$. So the solution set is a translate of an $(n-1)$-dimensional subspace. So, the solution set of $Ax = b$ is the intersection of 'm' such $(n-1)$-dimensional subspaces. **E.g.** if A is 3×3, the solution set is an intersection of 3 two-dimensional subspaces or planes.

Solving Linear Equations - Examples

Two examples from Treil (2024) (pages 40 - 45; section 2, chap 2).
How to derive a solution set as a translate of the null space of A.
This method also indicates a basis for $N(A)$.

Note(1) Let A be an $m \times n$ matrix, $b \in \mathbb{R}^m$ and E be an invertible $m \times m$ matrix.

- Let $S = \{x \in \mathbb{R}^n | Ax = b\}$, and $U = \{x \in \mathbb{R}^n | EAx = Eb\}$. Then $S = U$. Indeed let $x \in S$, so $Ax = b$.
- Premultiplying both sides preserves equality, so $EAx = Eb$, so $x \in U$.
- Conversely, let $x \in U$, so $EAx = Eb$. Premultiply both sides by E^{-1}, and we get $Ax = b$ or $x \in S$.

Solving Linear Equations - Examples

Note (2) Consider the system of equations $Ax = b$.
Solution steps -

- **(i)** Interchanging two rows,
- **(ii)** Multiplying a row by a scalar $a \neq 0$,
- **(iii)** replacing a row by its sum with a multiple of another row are operations that do not change the solutions to the system.

Each of these premultiplies $Ax = b$ with an invertible matrix E.

- E for interchanging rows j, k is the identity matrix I with rows j, k permuted.
- E for operation (ii) is I with row j multiplied by a.
- For (iii), $R_k \to R_k + aR_j$, E is I with the kj^{th} entry changed from 0 to a.

Solving Linear Equations

We use **Row Reduction**. Write down the augmented matrix $(A|b)$.

(1). Find leftmost non-zero column of the augmented matrix.

(2). Make sure, by row interchange if necessary, that the first entry of this column is non-zero. This is the *pivot*.

(3). Reduce all non-zero entries below the pivot by row operations of type 3: $R_j \to R_j + aR_1$.

(4). Ignore Row 1 and consider the matrix from Row 2 to Row m. Repeat steps (1) to (3). Iterate until the matrix is in **echelon form**. That is, the matrix has -

(i) all zero rows below non-zero rows, and

(ii) the first non-zero entry of a non-zero row is to the right of the first non-zero entry of the row above it.

(**iii**) Then use *back substitution* to solve the system.

Example: Consider the system $x_1 + 2x_2 + 3x_3 = 1$, $3x_1 + 2x_2 + x_3 = 7$, $2x_1 + x_2 + 2x_3 = 1$. The augmented matrix is:

Example 1: Unique Solution

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 3 & 2 & 1 & 7 \\ 2 & 1 & 2 & 1 \end{array}\right)$$

$R_2 - 3R_1$, $R_3 - 2R_1$ gives

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -4 & -8 & 4 \\ 0 & -3 & -4 & -1 \end{array}\right)$$

$R_3 - (3/4)R_2$ gives the **Echelon** form (lower triangular martix has 0's)–

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -4 & -8 & 4 \\ 0 & 0 & 2 & -4 \end{array}\right)$$

Back substitution gives $x_3 = -2, x_2 = 3, x_1 = 1$.

Q: What will happen if we have 0 in place of 2 in a_{33} position ?

Example 2: Multiple Solutions Example

Suppose the **Row reduced echelon form** (RREF) of a system is

$$\left(\begin{array}{ccccc|c} \boxed{1} & -2 & 0 & 0 & 0 & 1 \\ 0 & 0 & \boxed{1} & -5 & 0 & 2 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \end{array}\right)$$

(i) Identify the pivot (first nonzero entry) in each row (in boxed above);
(ii) Identify the columns without pivots (here, $2, 4$), thus variables x_2, x_4 are free - we can fix them at any real numbers and solve for x_1, x_3, x_5.
(iii) Write out the system of equations with the free variables on the RHS.

$$x_1 = 1 + 2x_2, \quad x_2 \text{ free}, \quad x_3 = 2 + 5x_4, \quad x_4 \text{ free}, \quad x_5 = 3.$$

(iv) Rewrite the solution: Stack them with $x \in \mathbb{R}^5$ on the left; on the right, there will be a 5×1 fixed vector plus x_2 multiplied by a 5×1 vector plus x_4 multiplied by a 5×1 vector.

$$\begin{aligned} (x_1, x_2, x_3, x_4, x_5)^T &= (1 + 2x_2,\ x_2,\ 2 + 5x_4,\ x_4,\ 3)^T \\ &= (1, 0, 2, 0, 3)^T + x_2(2, 1, 0, 0, 0)^T + x_4(0, 0, 5, 1, 0)^T \end{aligned}$$

Multiple Solutions

The soln set is a translate of subspace $span\{(2,1,0,0,0),(0,0,5,1,0)\}$.
So this span must be the Null Space of the RREF matrix U below

$$\begin{pmatrix} \boxed{1} & -2 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & -5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{pmatrix}$$

Note: $N(U) = \{x \in \mathbb{R}^5 | Ux = 0\}$; i.e. it's a solution set to the system of eqns $Ux = 0$. Since U is derived from the original matrix A in the system of eqns by repeated premultiplication with invertible elementary matrices, this solution set is the same as that of $Ax = 0$. i.e. $N(A) = N(U)$.
So, the solution set as written is of the form $x :=$ one particular solution to $Ax = b$, plus the Null Space of A.
In this example, the null space is a plane (2-dimensional) because $Rank(U) = 3$ (by the fundamental theorem of LA). And since $N(U) = N(A)$ equals $span(\{(2,1,0,0,0),(0,0,5,1,0)\})$, these vectors are a basis for it.

Representing a LT wrt standard bases

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be an LT. Let $\{e_1, ..., e_n\}$ be the standard basis for $\mathbb{R}^n$ [n-vectors], also let $\{e_1, ..., e_m\}$ be the standard basis for $\mathbb{R}^m$ [m-vectors].

Let any $x \in \mathbb{R}^n$, s.t. $x = \sum_1^n x_j e_j$. So, $T(x) = T(\sum_1^n x_j e_j) = \sum_1^n x_j T(e_j)$.

For $j = 1, ..., n$, let $\mathbf{a_j}$ be the m-vector of coordinates of $T(e_j)$ (i.e. representing $T(e_j)$ in terms of the standard basis $\{e_1, ..., e_m\}$ in $\mathbb{R}^m$). Then $T(x) = \sum_1^n x_j \mathbf{a_j} = Ax$, where $A = (\mathbf{a_1}, \mathbf{a_2}, ..., \mathbf{a_n})$.

We have 'represented' T as a matrix A, whose columns are $T(e_j), j = 1, ..., n$, the standard basis vectors of $\mathbb{R}^n$ mapped to $\mathbb{R}^m$ and represented in terms of the standard basis in $\mathbb{R}^m$. This is a general result.

Let V, W be vector spaces with bases $\mathcal{A} = \{a_1, ..., a_n\}$ and $\mathcal{B} = \{b_1, b_2, ..., b_m\}$ respectively. Suppose $v \in V$; in terms of **coordinates** with respect to basis $\mathcal{A}$, we'll call the vector $[v]_{\mathcal{A}}$. If $w \in W$, $[w]_{\mathcal{B}}$ has a similar meaning. We can show that there exists a matrix, denoted $[T]_{\mathcal{BA}}$, such that – $[w]_{\mathcal{B}} = [T(v)]_{\mathcal{B}} = [T]_{\mathcal{BA}}[v]_{\mathcal{A}}$.

Matrix Representation of LT

Theorem

Let $T : V \to W$; $\mathcal{A} = \{a_1, ..., a_n\}$, $\mathcal{B} = \{b_1, ..., b_m\}$ are bases in V, W respectively. Then there exists a matrix, denoted $[T]_{\mathcal{BA}}$ s.t. $w = T(v)$ implies $[w]_{\mathcal{B}} = [Tv]_{\mathcal{B}} = [T]_{\mathcal{BA}}[v]_{\mathcal{A}}$.

Proof. (not required).

Let $v = \sum_1^n x_j a_j$, i.e. coordinate representation $[v]_{\mathcal{A}} = (x_1, ..., x_n)^T$. So we have, $T(v) = T(\sum_1^n x_j a_j) = \sum_1^n x_j T(a_j)$, by linearity of T.
For $j = 1, ..., n$, now writing in terms of basis $\mathcal{B}$ (with i colms, $i = 1, ..., m$) –

$$T(a_j) = \sum_{i=1}^m a_{ij} b_i, \quad \text{i.e.} \quad [T(a_j)]_{\mathcal{B}} = (a_{1j}, ..., a_{mj})^T.$$

So, $$w = T(v) = \sum_{j=1}^n x_j \sum_{i=1}^m a_{ij} b_i = \sum_{i=1}^m b_i \sum_{j=1}^n a_{ij} x_j.$$

So if $[T(v)]_{\mathcal{B}} = (y_1, ..., y_m)^T$, we have, for $i = 1, ..., m$,
$y_i = \sum_{j=1}^n a_{ij} x_j$, so that $(y_1, ..., y_m)^T = A(x_1, ..., x_n)^T$, where the columns of A represent the $T(a_j)$'s w.r.t. $\mathcal{B}$. This $A \equiv [T]_{\mathcal{BA}}$. □

Matrix Representation - Applications

To summarize, if $w = T(v)$ in coordinate-free language, then if we write vectors in V and W in terms of bases $\mathcal{A}$ and $\mathcal{B}$ respectively, this mapping is achieved by the matrix $[T]_{\mathcal{B}\mathcal{A}}$. If a_j is the *jth* basis vector in $\mathcal{A}$, then the *jth* column of $[T]_{\mathcal{B}\mathcal{A}}$ is $[T(a_j)]_{\mathcal{B}}$, i.e. $T(a_j)$ written in terms of its coordinates w.r.t. the bases $\mathcal{B}$ of W.

By the balance of indices in the formula $[T(v)]_{\mathcal{B}} = [T]_{\mathcal{B}\mathcal{A}}[v]_{\mathcal{A}}$, Treil presumably means that to go from basis $\mathcal{A}$ to $\mathcal{B}$, the $\mathcal{B}$'s appear in the formula first, followed by the $\mathcal{A}$'s.

Change of Basis Matrix. As an application, consider a vector space V. Consider 2 different bases $\mathcal{A}$ and $\mathcal{B}$ for V. What is the mapping $[v]_{\mathcal{A}} \mapsto [v]_{\mathcal{B}}$? That is, suppose I first represent vectors v in terms of basis $\mathcal{A}$, and then want to find out their representation in terms of a different basis $\mathcal{B}$. Is there a matrix A s.t. $[v]_{\mathcal{B}} = A[v]_{\mathcal{A}}$? Replacing T in the above formula with I, we get $[I(v)]_{\mathcal{B}} = [I]_{\mathcal{B}\mathcal{A}}[v]_{\mathcal{A}}$, and note that $I(v) = v$.

Matrix Representation Applications

The *jth* column of $[I]_{\mathcal{BA}}$ is $[I(a_j)]_{\mathcal{B}} = [a_j]_{\mathcal{B}}$, the *jth* basis vector in $\mathcal{A}$ expressed in terms of the basis $\mathcal{B}$. $[I]_{\mathcal{BA}}$ is called the **change of basis matrix**. Premultiplying a vector written in terms of coordinates wrt basis $\mathcal{A}$ yields the same vector written in terms of coordinates wrt basis $\mathcal{B}$.

Example

In $\mathbb{R}^2$, change basis from the standard basis to $\{(1,2),(2,1)\}$. So we express the standard basis vectors in terms of the new basis vectors. $(1,0)^T = a_{11}(1,2)^T + a_{21}(2,1)^T$ implies $(a_{11}, a_{21})^T = B^{-1}(1,0)^T$, where

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Similarly, $(a_{12}, a_{22})^T = B^{-1}(0,1)^T$. So the change of basis matrix is $B^{-1}I = B^{-1}$. So generally, *any* full rank $n \times n$ matrix A can be regarded as a change of basis matrix.

Application - Composition of LTs

Lemma

Let $T_1 : X \to Y$, $T_2 : Y \to Z$ be LTs, and let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be respective bases for X, Y, Z. Let $T = T_2 \circ T_1 : X \to Z$. Then
$[T]_{\mathcal{C}\mathcal{A}} \equiv [T_2 \circ T_1]_{\mathcal{C}\mathcal{A}} = [T_2]_{\mathcal{C}\mathcal{B}}[T_1]_{\mathcal{B}\mathcal{A}}$.

Proof. (not required).

On the LHS, the *jth* column of $[T]_{\mathcal{C}\mathcal{A}}$ is $[T(a_j)]_{\mathcal{C}}$, where a_j is the *jth* basis vector in $\mathcal{A}$. (This follows from $[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}\mathcal{A}}[v]_{\mathcal{A}}$).

On the RHS, the *jth* column of $[T_2]_{\mathcal{C}\mathcal{B}}[T_1]_{\mathcal{B}\mathcal{A}}$ is $[T_2]_{\mathcal{C}\mathcal{B}}[T_1(a_j)]_{\mathcal{B}}$, where the latter vector is the jth column of $[T_1]_{\mathcal{B}\mathcal{A}}$. But then we have $[T_2(T_1(a_j))]_{\mathcal{C}} = [T_2]_{\mathcal{C}\mathcal{B}}[T_1(a_j)]_{\mathcal{B}}$.
And the LHS of this equals $[T(a_j)]_{\mathcal{C}}$, since $T(x) = T_2(T_1(x)), \forall x$. □

Note: Lemma generalizes the notion that matrix multiplication is akin to function composition. If A, B are respectively $m \times n$ and $n \times p$ matrices, then the $m \times p$ product $AB : \mathbb{R}^p \to \mathbb{R}^m$.

Change of Basis and Similarity

(1). A change of basis matrix $[I]_{\mathcal{BA}}$ is invertible because its columns are LI, being basis vectors. Its inverse is $[I]_{\mathcal{AB}}$, since $[I]_{\mathcal{AB}}[I]_{\mathcal{BA}} = I$.

$$\underleftarrow{[I]_{\mathcal{AB}}} \cdot \underleftarrow{[I]_{\mathcal{BA}}} \cdot [v]_{\mathcal{A}} = [I]_{\mathcal{AB}}[v]_{\mathcal{B}} = [v]_{\mathcal{A}}$$

(2). Suppose we know the matrix $[T]_{\mathcal{BA}}$ that represents the LT T wrt bases $\mathcal{A}, \mathcal{B}$, and wish to express T wrt different bases $\mathcal{A}', \mathcal{B}'$. The relevant matrix is derived by the formula:

$$[T]_{\mathcal{B}'\mathcal{A}'} = [I]_{\mathcal{B}'\mathcal{B}} \cdot [T]_{\mathcal{BA}} \cdot [I]_{\mathcal{AA}'}$$

That is, if we postmultiply the RHS with a vector v in the domain expressed as $[v]_{\mathcal{A}'}$, first the vector is expressed in terms of basis $\mathcal{A}$, then this is carried by $[T]_{\mathcal{BA}}$ to the codomain and expressed in terms of basis $\mathcal{B}$, and finally this is expressed in terms of basis $\mathcal{B}'$.

Change of Basis and Similarity

(3). If $T : V \to V$, let $[T]_{\mathcal{A}\mathcal{A}}$ refer to the matrix representation if the basis is $\mathcal{A}$ in both domain and codomain.
For 2 bases $\mathcal{A}, \mathcal{B}$, we therefore have

$$[T]_{\mathcal{B}\mathcal{B}} = [I]_{\mathcal{B}\mathcal{A}} \cdot [T]_{\mathcal{A}\mathcal{A}} \cdot [I]_{\mathcal{A}\mathcal{B}} \;=\; [I]^{-1}_{\mathcal{A}\mathcal{B}} \cdot [T]_{\mathcal{A}\mathcal{A}} \cdot [I]_{\mathcal{A}\mathcal{B}}.$$

Definition (Similar Matrices)

Thus we say two matrices P, Q are **similar** if there exists an invertible matrix S s.t.

$$Q = S^{-1} \cdot P \cdot S$$

P and Q thus 'represent' the same LT w.r.t. different bases.